ZHOU, CONGHAO

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CONTACT

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RESEARCH INTERESTS

dark matter · gravitational lensing · galaxy clusters · large scale structure
shear calibration · image simulation

EDUCATION

University of California, Santa Cruz

Expected 2027 Ph.D. in Physics
Physics Department
Thesis: *TBD*
Comitee: Tesla E. Jeltema, Alexie S. Leauthaud

Duke University

2020 B.A. in Physics with Highest Distinction
Department of Physics
Thesis: *Intrinsic Alignment of redMaPPer Clusters in the Dark Energy Survey*
Comitee: Michael A. Troxel, Christopher W. Walter, Robert L. Wolpert

SCIENCE PUBLICATIONS WITH MAJOR CONTRIBUTIONS

1. Constraining optical selection effection with DES and SPT, in prep.

   C. Zhou, S. Grandis, H.Y. Wu, A.N. Salcedo et al.

2. Resposne Bias in Shear Calibration, in prep.

   C. Zhou, M.R. Becker, M. Gatti et al.

3. Accurate Galaxy Cluster Shear and Mass Calibration for LSST with AnaCal, in collaboration review

   C. Zhou, X. Li et al.

4. Relationship between 2D and 3D Galaxy Stellar Mass and Correlations with Halo Mass, JCAP

   C. Zhou, A.S. Leauthaud, S. Xu., B. Diemer, S. Huang, et al.

5. Forecasting the constraints on optical selection bias and projection effects of galaxy cluster lensing with multiwavelength data, PRD Editors' Suggestion

   C. Zhou, H.Y Wu, A.N. Salcedo, S. Grandis, T.E. Jeltema, A.S. Leauthaud et al.

6. The intrinsic alignment of red galaxies in DES Y1 redMaPPer galaxy clusters ,

   MNRAS
   C. Zhou, A. Tong, M.A. Troxel, J. Blazek, C. Lin et al.

TECHNICAL PUBLICATIONS WITH MAJOR CONTRIBUTIONS

1. Surface brightness profiles around massive galaxies in LSSTComCam data ,

   LSST Technote
   C. Zhou, T. Jeltema, A. von der Linden, L.S. Kelvin et al.

CONTRIBUTING AUTHOR SCIENCE PUBLICATIONS

1. The Vera C. Rubin Observatory Data Preview 1 , arXiv

   Vera C Rubin Observatory Team.

2. Impact of projection-induced optical selection bias on the weak lensing mass calibration of galaxy clusters , PRD

   T.N. Nde et al.

3. DESI-DR1 3×2-pt analysis: consistent cosmology across weak lensing surveys , arXiv

   A. Porredon et al.

4. Reaching for the Edge II: Stellar Halos out to Large Radii as a Tracer of Dark Matter Halo Mass , arXiv

   K. Leidig et al.

5. Cosmological Constraints from Dark Energy Survey Year 1 Cluster Lensing and Abundances with Simulation-based Forward-Modeling , arXiv

   A.N. Salcedo et al.

6. Testing the Stellar Feedback-driven Breathing Mode in Low-mass Galaxies with Gas Kinematics , arXiv

   Y. Luo et al.

7. Dark energy survey year 3 results: Cosmological constraints from cluster abundances, weak lensing, and galaxy clustering , PRD

   T.M.C. Abbott et al.

8. NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1) , arXiv

   C.O. Chandler et al.

9. Data Release 1 of the Dark Energy Spectroscopic Instrument , arXiv

   DESI Collaboration

10. Association between optically identified galaxy clusters and the underlying dark matter halos , PRD

    S. Cao et al.

11. The Outskirt Stellar Mass of Low-Redshift Massive Galaxies is an Excellent Halo Mass Proxy in Illustris/IllustrisTNG Simulations , ApJ

    S. Xu et al.

12. Improving Galaxy Cluster Selection with the Outskirt Stellar Mass of Galaxies , PRD

    M.Kwiecien et al.

13. Optical galaxy cluster mock catalogs with realistic projection effects: validations with the SDSS redMaPPer clusters , PRD

    A. Lee et al.

14. Building an Efficient Cluster Cosmology Software Package for Modeling Cluster Counts and Lensing , arXiv

    M. Aguena et al.

15. Dark Energy Survey Year 3 results: Simulation-based cosmological inference with wavelet harmonics, scattering transforms, and moments of weak lensing mass maps. Validation on simulations , PRD

    M. Gatti et al.

16. Dark Energy Survey Year 3 Results: Deep Field Optical + Near-Infrared Images and Catalogue , MNRAS

W.G. Hartley et al.

talks

dec. 2024	Theoretical Astrophysics and Cosmology Seminar, University of Arizona
nov. 2024	Cosmology Seminar, UC Davis
apr. 2024	CMB S4 Cluster AWG Meeting
mar. 2024	Center for the Fundamental Physics of the Universe Seminar, Brown University
mar. 2024	Journal Club, University of Pennsylvania
oct. 2023	Cosmology Seminar, Tsinghua University
oct. 2023	Joint Stony Brook University/ Brookhaven National Lab Cosmology Seminar

observing experience

10.5 effective nights	Dark Energy Spectroscopic Instrument, Mayall 4m at Kitt Peak
3 nights	DECaLS Intermediate Band Imaging Survey, Blanco 4m at CTIO

professional membership

2026 - present	Member, Roman Science Collaboration
2024 - present	In-Kind Contributor, Rubin Observatory Commissioning
2024 - Present	Full Member, LSST Dark Energy Science Collaboration
2024 - Present	Student Member, American Astronomical Society
2024 - Present	Student Member, American Physical Society
2023 - Present	Member, LSST Galaxy Science Collaboration
2023 - Present	Member, LSST Informatics & Statistics Science Collaboration
2020 - Present	Sponsored Member, Dark Energy Spectroscopic Instrument
2020 - 2024	Member, LSST Dark Energy Science Collaboration
2018 - Present	Student Member, Dark Energy Survey

outreach

2023 - 2024	Starlight Over Street Light
2022	UC Santa Cruz Physics Outreach Taskforce
2022	Science Internship Program Mentor
2022	UC Santa Cruz Physics Mentoring Program Mentor

professional service

2025 Winter UC Santa Cruz Physics Colloquium Committee

2024 - present Referee, Astronomy & Astrophysics

jul.2023-jun.2024 Organizer, BCG Seminar, Boise State

sep.2023-jun.2024 Organizer, CosmoGal Seminar, UC Santa Cruz

jan. 2023 LOC, APS Conference for Undergraduate Women in Physics at UC Santa Cruz

aug. 2022 SOC, LSST DESC 2022 Chicago Meeting

apr.2022-may.2025 DES Early Career Scientists Committee

jul. 2022 - present LSST DESC Mentoring Comittee

TEACHING EXPERIENCE

2024 Spring Application of Quantum Materials (Grader)

2024 Winter Electricity, Magnetism, and Optics (TA)

2022 Fall Quantum Mechanics (TA)

2022 Summer Intermediate Physics Laboratory (TA)

2022 Spring Advanced Astrophysics Laboratory (TA)

3 times Introductory Physics Laboratory (TA)

April 14, 2026